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Bridging machine learning and clinical practice: a multicentre nomogram for 90-day graft failure risk stratification in heart transplantation

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ABSTRACT

Background Early graft failure within 90 postoperative days is the leading cause of mortality after heart transplantation. Existing risk scores, based on linear regression, often struggle to capture the complex, multifactorial biological interactions necessary for personalised donor–recipient matching. This study utilised explainable machine learning (ML) to identify robust predictors of 90-day graft failure and developed a clinically interpretable, ML-informed nomogram designed specifically for cross-population generalisability.

Methods Using the UNOS registry (2008–2020; n=25 200), XGBoost/Random Forest models identified 90-day graft failure predictors from 32 donor–recipient variables. Explainable AI (SHapley Additive exPlanations) analysis revealed key predictors and their non-linear interactions, which were translated into a clinically applicable nomogram. External validation was performed on a large, single-centre Chinese cohort (Wuhan Union Hospital ; 2018–2023; n=563), assessing performance via area under the curve (AUC), calibration and decision curve analysis (DCA).

Findings The final model incorporated eight predictors: recipient factors (prior cardiac surgery, age, bilirubin, body mass index (BMI)), donor factors (age, gender, BMI) and cold ischaemia time. The XGBoost-derived nomogram demonstrated consistent discrimination (AUC 0.67, 95% CI 0.64 to 0.70) and calibration. Patients stratified into the high-risk group (top quantile by nomogram score) had a 2.4-fold increased hazard of graft failure (HR 2.42, 95% CI 2.11 to 2.78). DCA confirmed the model's clinical utility across a wide range of risk thresholds (0.0–0.4). External validation in the Chinese cohort affirmed its generalisability (AUC 0.67).

Conclusion This study introduces an ML-informed nomogram for 90-day graft failure, validated across USA and Chinese populations. By translating ML insights into a clinically interpretable tool using routinely available pretransplant variables, it bridges a key translational gap in transplant risk prediction. This tool can aid in optimising donor–recipient matching and personalising post-transplant management, with the potential to help address geographic disparities in heart transplant outcomes.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Early graft failure within 90 days is the primary cause of mortality after heart transplantation, yet existing risk scores like Index for Mortality Prediction After Cardiac Transplantation (IMPACT) score rely on linear models that fail to capture complex, non-linear donor–recipient interactions. While machine learning (ML) can decipher these intricate relationships, its 'black-box' nature has hindered clinical adoption in heart transplantation.

WHAT THIS STUDY ADDS

⇒ Using the UNOS registry and Explainable AI (SHapley Additive exPlanations; SHAP), we identified eight robust pretransplant predictors—including recipient bilirubin, prior surgery and cold ischaemia time—and translated them into a clinically interpretable, linear nomogram. This tool maintains the predictive power of complex ML models (area under the curve 0.67) while providing a transparent, bedside-ready format for risk assessment.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Validated across both USA and Chinese cohorts, the nomogram offers a practical, cross-continental aid for optimising donor–recipient matching and personalising postoperative management. This approach demonstrates a pipeline for bridging the translational gap between high-dimensional artificial intelligence (AI) discovery and actionable clinical decision support.

INTRODUCTION

Heart transplantation (HTx) is the definitive therapy for end-stage heart failure, yet early graft failure (EGF) within 90 days remains the principal cause of postoperative mortality, accounting for 30%–40% of deaths in this critical period.[1 2] EGF, often stemming from primary graft dysfunction, rejection or infection, underscores the urgent need for robust predictive tools to optimise donor–recipient matching and guide post-operative care.[3 4]

Existing risk scores, such as Index for Mortality Prediction After Cardiac Transplantation (IMPACT) and Eurotransplant, rely on traditional linear regression models and static variables, offering only moderate predictive accuracy.[5–8] Their primary limitation is an inability to capture the complex, non-linear interactions between donor, recipient and procedural factors that collectively determine graft survival.[9–11] This gap highlights the demand for more dynamic and personalised risk assessment tools.

Machine learning (ML) offers transformative potential by deciphering these intricate, multifactorial relationships—such as how donor–recipient BMI mismatch might exacerbate the risks of prolonged cold ischaemia time (CIT).[12–14] However, the 'black-box' nature of many ML algorithms has hindered their clinical adoption. The emergence of explainable AI techniques, like SHapley Additive exPlanations (SHAP), now enables the translation of complex model outputs into clinically interpretable insights, bridging the gap between algorithmic power and actionable clinical guidance.[15 16] While such approaches have advanced fields like liver transplantation, a parallel tool for HTx is conspicuously absent.[17 18]

In this study, we leveraged a large international registry to develop an ML model for predicting 90-day graft failure. We then applied SHAP analysis to identify and interpret the key drivers of risk, transforming these findings into a clinically practical nomogram. This tool was rigorously validated using an independent cohort from a high-volume Chinese centre, assessing its generalisability across diverse healthcare settings.[10 19] By uniting ML's predictive accuracy with clinical interpretability, our work aims to provide a practical solution for personalising transplant care and improving early outcomes. This retrospective study was reported in accordance with the Strengthening the reporting of cohort, cross-sectional and case-control studies in surgery (STROCSS) guidelines.[20]

MATERIALS AND METHODS

Comprehensive descriptions of the materials and methods are available in online supplemental data. This includes details on the data source and study population, the follow-up protocol and definitions of key variables and outcomes. The methodology for ML encompassed data preprocessing, dataset splitting and model development. Model interpretability was assessed using SHAP, and a nomogram was developed for clinical utility. The section also outlines the statistical analyses performed and addresses ethical considerations.

RESULTS

Study population, design and data flow

We followed a multicentre, retrospective cohort study design (figure 1). The derivation cohort consisted of adult HTx recipients from the UNOS database (2008–2020), and the external validation cohort included patients from Wuhan Union Hospital (WHUH) (2015–2024). Information on variables with missing data is shown in online supplemental table S1. After applying inclusion/exclusion criteria and data preprocessing (eg, handling missing values, normalisation), we retained 25 200 and 563 patients from UNOS and WHUH for analysis. The donor and recipient's baseline characteristics with and without 90 days graft failures are shown in tables 1 and 2, respectively. The baseline characteristics for the training and validation sets for UNOS data are shown in online supplemental table S2.

Model performance

Both XGBoost and random forest (RF) demonstrated moderate predictive performance for 90-day graft

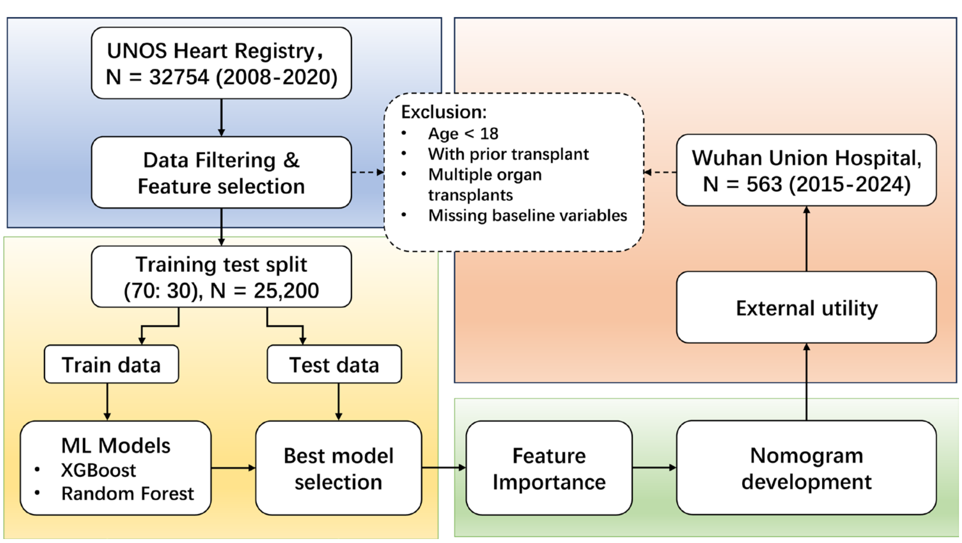

Figure 1 Flowchart of model development and validation. This multicentre retrospective cohort study outlines the machine learning pipeline for predicting 90-day graft failure. Data were sourced from the UNOS Heart Registry (N=32 754; 2008–2020) and Wuhan Union Hospital (N=563; 2015–2020). UNOS, United Network for Organ Sharing.

Table 1 Donor and recipient baseline characteristics with and without graft failure at 90 days post-transplantation

	Non-failure, N=23 756*	Graft failure, N=1444*	P value†
Donor variables			
Age	30 (22, 40)	32 (23, 44)	<0.001
Male gender	16 814 (71)	950 (66)	<0.001
BMI (kg/m^2)	26.4 (23.3, 30.4)	26.6 (23.5, 30.2)	0.6
Medication use			
Inotropic support	10 283 (43)	662 (46)	0.063
Arginine vasopressor	15 988 (67)	971 (67)	>0.9
Antihypertensive	7322 (31)	417 (29)	0.12
Cause of death			<0.001
Anoxia	6902 (29)	383 (27)	
CNS tumour	123 (0.5)	12 (0.8)	
CVS	4386 (18)	336 (23)	
Head trauma	11 814 (50)	671 (46)	
Other	531 (2.2)	42 (2.9)	
LVEF (%)	60 (55, 65)	60 (55, 65)	0.4
Medical history			
Hypertension	3633 (15)	256 (18)	0.013
Diabetes	851 (3.6)	52 (3.6)	>0.9
Alcoholism	3871 (17)	264 (19)	0.056
Smoking	2909 (12)	202 (14)	0.053
Creatinine	1.00 (0.80, 1.40)	1.00 (0.78, 1.40)	0.4
BUN	16 (11, 25)	16 (11, 24)	0.10
CIT (hour)	3.20 (2.40, 3.80)	3.40 (2.70, 4.10)	<0.001
Recipient factors			
Age	56 (46, 63)	58 (50, 64)	<0.001
Gender	17 490 (74)	1072 (74)	0.6
BMI (kg/m^2)	27.1 (23.8, 30.8)	27.9 (24.3, 31.6)	>0.9
Main diagnosis			<0.001
CAD	759 (3.2)	57 (3.9)	
CHD	731 (3.1)	88 (6.1)	
DCM	19 782 (83)	1170 (81)	
HCM	627 (2.6)	28 (1.9)	
RCM	867 (3.6)	53 (3.7)	
VHD	342 (1.4)	20 (1.4)	
Other	648 (2.7)	28 (1.9)	
Previous cardiac surgery	9235 (39)	751 (53)	<0.001
Total HLA mismatch levels	5.00 (4.00, 5.00)	5.00 (4.00, 5.00)	0.9
HLA-DR mismatch levels			
0	1076 (5.0)	63 (4.9)	

Continued

Table 1 Continued

	Non-failure, N=23 756*	Graft failure, N=1444*	P value†
1	8677 (40)	532 (41)	
2	11 834 (55)	703 (54)	
Cigarette use	11 011 (46)	713 (49)	0.025
Total bilirubin	0.70 (0.50, 1.10)	0.82 (0.50, 1.40)	<0.001
Inotropes use	9010 (38)	490 (34)	0.003
ECMO at listing	254 (1.1)	43 (3.0)	<0.001
LVAD at listing	4354 (18)	304 (21)	0.011
Centre volume			0.003
Low	665 (2.8)	41 (2.8)	
Medium	11 144 (47)	743 (51)	
High	11 947 (50)	660 (46)	
Gender mismatch	5698 (24)	390 (27)	0.010
Race mismatch	18 685 (79)	1173 (81)	0.022

*Median (IQR); n (%).
†One-way ANOVA; Pearson's χ^2 test.
BMI, body mass index; BUN, blood urea nitrogen; CAD, coronary artery disease; CHD, congenital heart disease; CIT, cold ischaemia time; CNS, central nervous system; CVS, cerebrovascular/stroke; DCM, dilated cardiomyopathy; DR, human leukocyte antigen (HLA)-DR; ECMO, extracorporeal membrane oxygenation; HCM, hypertrophic cardiomyopathy; HLA, human leukocytic antigen; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; RCM, restrictive cardiomyopathy; VHD, valvular heart disease.

failure, with XGBoost achieving a marginally higher area under the curve (AUC) (0.6692, 95% CI 0.6355 to 0.703) compared with RF (95% CI 0.6257 to 0.6965) (figure 2A). While the difference in discrimination was small, XGBoost was selected for downstream analysis and nomogram development due to a calibration superiority, with a Brier score of 0.0471 versus 0.0474 for RF. Both models were optimised using fivefold cross-validation, with hyperparameters tuned to prioritise clinical utility over marginal performance gains (eg, avoiding overfitting to rare subgroups). Notably, in a head-to-head comparison using the UNOS dataset, the traditional IMPACT score demonstrated lower discriminative capacity for EGF outcomes, yielding an AUC of 0.586 (online supplemental figure S1).

Variable importance and clinical relevance

To identify stable predictors from 32 candidate variables, SHAP analysis (figure 2B) identified the top eight predictors based on their global importance, calculated as the mean absolute SHAP values across the training cohort, including prior cardiac surgery, total bilirubin (mg/dL), recipient age and CIT (hour) as the strongest predictors of 90-day graft failure, followed by donor age (year), BMI (kg/m^2), gender (male), centre volume, donor

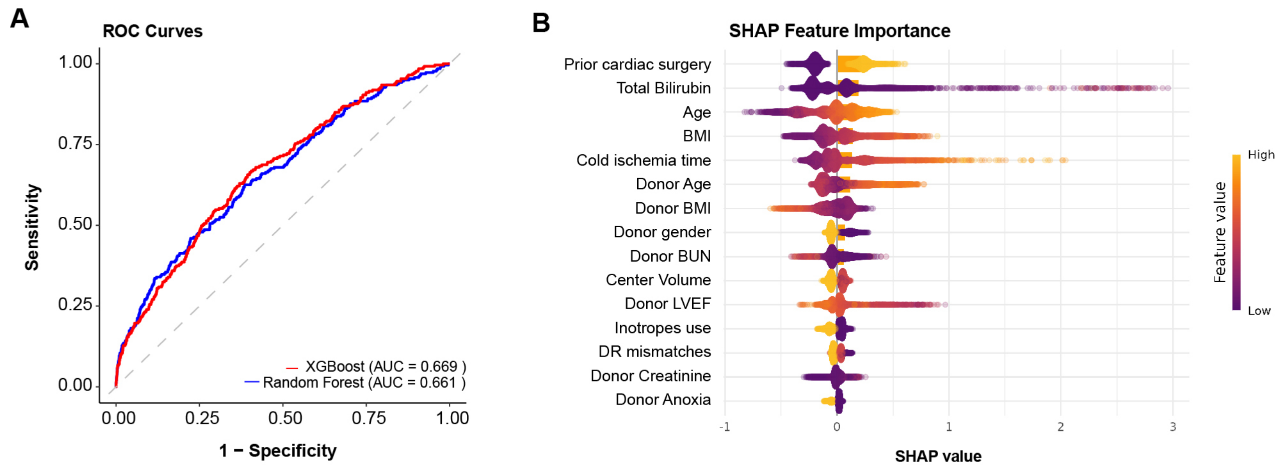

Figure 2 Model performance and feature importance analysis. (A) Receiver operating characteristic (ROC) curves comparing XGBoost (red) and Random Forest (blue) models for predicting 90-day graft failure. Dotted diagonal line represents no discrimination reference. (B) SHapley Additive exPlanations (SHAP) summary plot using the TreeExplainer estimator to ensure exact additive feature attributions. The plot integrates feature importance (bar plot) and directional impact (beeswarm). Features are ranked by mean absolute SHAP value (mean(|SHAP|)) representing their average impact on the model's log-odds prediction. Yellow/purple dots in the beeswarm plot indicate positive/negative SHAP values, reflecting risk-enhancing or protective effects. AUC, area under the curve; BMI, body mass index; BUN, blood urea nitrogen; LVEF, left ventricular ejection fraction; DR, Human Leukocyte Antigen (HLA)-DR.

left ventricular ejection fraction (%), donor inotropes use, Human Leukocyte Antigen (HLA)DR mismatches, donor creatinine and donor anoxia.

Clustering analysis and risk group stratification

To explore heterogeneity in risk patterns, we performed K-means clustering (K=3) on SHAP values from the training cohort. Principal component analysis revealed distinct clusters in reduced dimensional space (figure 3A), suggesting subgroups with divergent risk profiles. Baseline characteristics of 3 SHAP clusters are shown in table 2. The moderate-risk cluster (n=7612) consists of recipients with a history for prior cardiac surgery ($p<0.001$), while the high-risk cluster (n=1367) was dominated by pretransplant morbidity markers including elevated recipient bilirubin ($p<0.001$) as well as increased age ($p<0.001$) and donor factors including CIT ($p<0.001$) (figure 2B; table 2). The low-risk cluster ($n = 11{,}181$) showed minimal contributions from adverse predictors. Cumulative hazard curves (figure 3C) confirmed significant differences in 90-day graft failure rates across clusters (log-rank $p<0.001$), with the high-risk cluster accounting for 168 of events, survival rate of 87.7%, moderate-risk cluster accounting for 539 of events, survival rate of 92.9% and low-risk cluster accounting for

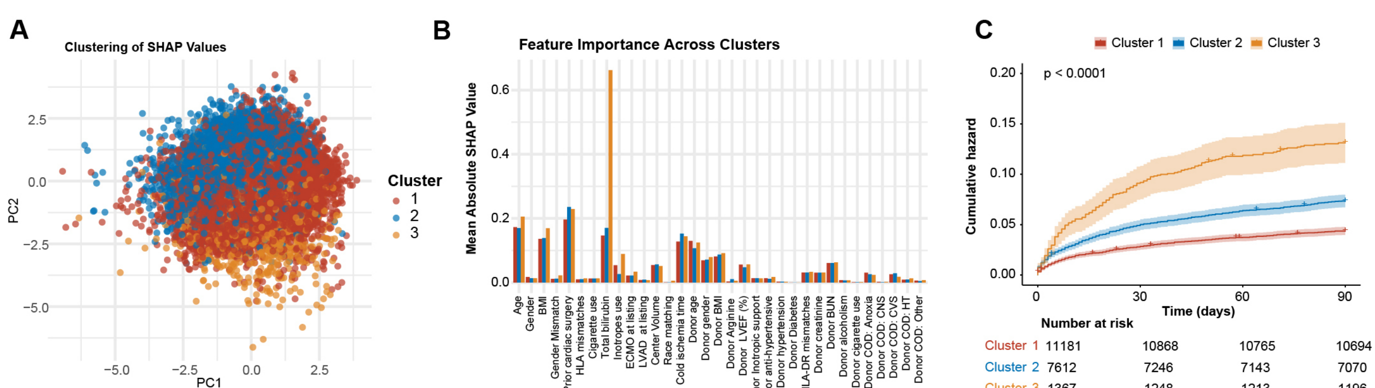

Figure 3 Unsupervised phenotyping via SHAP value clustering. (A) Principal component analysis (PCA) plot of high-dimensional SHAP contribution matrix utilised to visualise the functional separation of risk phenotypes in a reduced 2D space, stratified by K-means cluster (K=3). (B) Barplot showing mean absolute SHAP values calculated independently for each identified cluster, revealing divergent mechanistic drivers of risk across subpopulations. (C) Cumulative hazard curves stratified by clusters confirmed divergent outcomes (log-rank test). Cluster 1 (red) was identified as lowest-risk, followed by cluster 2 (blue) as moderate-risk and cluster 3 (orange) as highest-risk group. BMI, body mass index; BUN, blood urea nitrogen; CNS, central nervous system; CVS, cerebrovascular/stroke; ECMO, extracorporeal membrane oxygenation; HLA, human leukocytic antigen; LVEF, left ventricular ejection fraction; SHAP, SHapley Additive exPlanations. HT, head trauma;COD,cause of death; DR, Human Leukocyte Antigen (HLA)-DR;LVAD,left ventricular assist device; PC, principal components.

Table 2 Donor and recipient baseline characteristics of three SHAP clusters

	C1, N=11 181*	C2, N=7612*	C3, N=1367*	P value†
Donor variables				
Age	30 (23, 40)	30 (22, 40)	31 (23, 42)	0.10
Gender	7752 (69)	5471 (72)	973 (71)	<0.001
BMI (kg/m^2)	26.3 (23.1, 30.4)	26.4 (23.4, 30.3)	26.4 (23.4, 30.3)	>0.9
Medication use				
Inotropic support	4921 (44)	3276 (43)	593 (43)	0.4
Arginine vasopressor	7553 (68)	5074 (67)	908 (66)	0.4
Antihypertensive	3420 (31)	2359 (31)	399 (29)	0.4
Cause of death				0.031
Anoxia	3184 (28)	2251 (30)	345 (25)	
CNS tumour	62 (0.6)	36 (0.5)	12 (0.9)	
CVS	2075 (19)	1412 (19)	287 (21)	
Head trauma	5602 (50)	3736 (49)	688 (50)	
Other	258 (2.3)	177 (2.3)	35 (2.6)	
LVEF (%)	60 (55, 65)	60 (55, 65)	60 (55, 65)	>0.9
Medical history				
Hypertension	1676 (15)	1247 (16)	204 (15)	0.029
Diabetes	404 (3.6)	273 (3.6)	36 (2.6)	0.2
Alcoholism	1856 (17)	1317 (17)	230 (17)	0.5
Smoking	1378 (12)	969 (13)	205 (15)	0.019
CIT (hour)	1.00 (0.80, 1.40)	1.00 (0.80, 1.40)	1.00 (0.80, 1.40)	0.4
BUN (mg/dL)	16 (11, 25)	16 (11, 25)	16 (11, 24)	0.11
Recipient factors				
Age	55 (46, 62)	58 (49, 64)	54 (42, 62)	<0.001
Gender	7931 (71)	5885 (77)	1040 (76)	<0.001
BMI (kg/m^2)	27.0 (23.7, 30.8)	27.6 (24.2, 31.1)	26.1 (23.0, 29.8)	<0.001
Main diagnosis				<0.001
CAD	249 (2.2)	359 (4.7)	42 (3.1)	249 (2.2)
CHD	76 (0.7)	511 (6.7)	57 (4.2)	76 (0.7)
DCM	9546 (85)	6115 (80)	1123 (82)	9546 (85)
HCM	363 (3.2)	130 (1.7)	32 (2.3)	363 (3.2)
RCM	575 (5.1)	112 (1.5)	53 (3.9)	
VHD	40 (0.4)	222 (2.9)	24 (1.8)	
Other	332 (3.0)	163 (2.1)	36 (2.6)	
Previous cardiac surgery	0 (0)	7612 (100)	490 (36)	<0.001
HLA mismatch levels	5.00 (4.00, 5.00)	5.00 (4.00, 5.00)	5.00 (4.00, 5.00)	0.047
HLA-DR mismatch levels				0.7
0	532 (4.8)	397 (5.2)	72 (5.3)	
1	4506 (40)	3069 (40)	554 (41)	
2	6143 (55)	4146 (54)	741 (54)	
Cigarette use	4774 (43)	4039 (53)	573 (42)	<0.001
Total bilirubin	0.70 (0.50, 1.00)	0.70 (0.50, 1.00)	2.90 (2.30, 3.80)	<0.001
Inotropes use	4592 (41)	2338 (31)	664 (49)	<0.001
ECMO at listing	119 (1.1)	71 (0.9)	56 (4.1)	<0.001
LVAD at listing	1213 (11)	2298 (30)	187 (14)	<0.001

Continued

Table 2 Continued

	C1, N=11 181*	C2, N=7612*	C3, N=1367*	P value†
Centre volume				0.5
Low	294 (2.6)	224 (2.9)	46 (3.4)	
Medium	5292 (47)	3566 (47)	637 (47)	
High	5595 (50)	3822 (50)	684 (50)	
Race mismatch	8492 (76)	6311 (83)	1054 (77)	<0.001

*Median (IQR); n (%).
†One-way ANOVA; Pearson's χ^2 test.
BMI, body mass index; BUN, blood urea nitrogen; CAD, coronary artery disease; CHD, congenital heart disease; CIT, cold ischaemia time; CNS, central nervous system; CVS, cerebrovascular/stroke; DCM, dilated cardiomyopathy; ECMO, extracorporeal membrane oxygenation; HCM, hypertrophic cardiomyopathy; HLA, human leukocytic antigen; LVEF, left ventricular ejection fraction; RCM, restrictive cardiomyopathy; VHD, valvular heart disease.

478 of events, survival rate of 95.7%. This analysis justifies the need for personalised risk stratification, as uniform thresholds may overlook critical subgroup interactions.

Individual SHAP explanation

To illustrate how variables influence patient-specific risk, we generated SHAP force plots for representative cases across risk clusters: low-risk, moderate-risk and high-risk cluster (figure 4). In the lowest risk patient, features here predominantly show negative SHAP values (purple bars), meaning they reduce the predicted risk of graft failure. In the middle to high-risk individual, features here show positive SHAP values (yellow bars), indicating they increase the predicted risk. Donor-related features like donor age=51 (SHAP+0.216) and CIT=4.3 (SHAP+0.231) as well as recipient peri-operative features such as prior cardiac surgery=1 (SHAP+0.214) and age=66 (SHAP+0.265) are top contributors to graft failure in the highest risk cluster. In the moderate risk individual, in the moderate risk cluster, despite donor BMI=27.4 (SHAP+0.135), prior cardiac surgery=1 (SHAP+0.24) and total bilirubin of 1.1 mg/dL (SHAP+0.0927), other features contributed towards decreased graft failure risk, such as CIT=1.6 (SHAP −0.226) and donor age=32 (SHAP −0.121). These examples underscore the model's capacity to disentangle multifactorial risk, offering clinicians granular insights for individualised pretransplant counselling and postoperative monitoring.

External validation using machine learning-guided nomogram

While these interactions were informative, our primary goal was a bedside tool. Therefore, we translated the discovery phase output into a user-friendly nomogram (figure 5) using the high-importance variables, defined as the eight predictors with the highest global importance, calculated as the mean absolute SHAP values across the training cohort derived from the ML model. While standard variable selection methods like bootstrap inclusion were considered, the SHAP-based approach was prioritised for its ability to reflect feature impact within the context of the ensemble model. The translation of the XGBoost model into a simpler nomogram entailed a minimal, non-significant loss in discrimination (ΔAUC=0.0049; p=0.644 by DeLong's test; online supplemental figure S2) and slightly better calibration (Brier score: 0.0472 vs 0.0476). This indicates that the linear combination of these ML-identified features captures the dominant predictive signal for 90-day graft failure, confirming that the primary predictive signal identified by the ML discovery phase was effectively captured by the final additive model. The nomogram was then validated in the WHUH cohort (n=563). It demonstrated moderate discrimination for 90-day graft failure, achieving an AUC of 0.671 (95% CI 0.6025 to 0.7386) (figure 6A), which closely mirrored the performance of the XGBoost model in the derivation cohort. Decision curve analysis (figure 6B) confirmed its clinical utility, showing net benefit over treat-all or treat-none strategies at risk thresholds between 0.0 and 0.4, suggesting relevance for guiding interventions in patients with intermediate-to-high predicted risk. This indicates that the nomogram strategy is the most beneficial when considering the balance between the costs and benefits of interventions at these thresholds. To show the disparities of HTx recipients at distinct risk, we stratified the cohort into low-risk (nomogram score <30) and high-risk (nomogram score ≥30) groups using the nomogram. Cumulative hazard curves revealed stark differences in 90-day graft failure rates, with the high-risk group (n=281) exhibiting a cumulative hazard rate of 17.14% (95% CI 12.61% to 21.44%), compared with 7.45% (95% CI 4.34% to 10.47%) in the low-risk group (n=282) (figure 6C). Baseline characteristics of the low and high-risk groups are summarized in table 3. Cox proportional hazard model indicates that high-risk group shows more than two times the risk compared with the low-risk group (HR 2.42, 95% CI 2.11 to 2.78; p<0.001) (figure 6D), underscoring the nomogram's ability to effectively stratify patients into clinically meaningful risk categories.

Occurrence of post-transplant adverse events

Post-transplant adverse events were compared between low-risk and high-risk groups (table 4). The high-risk

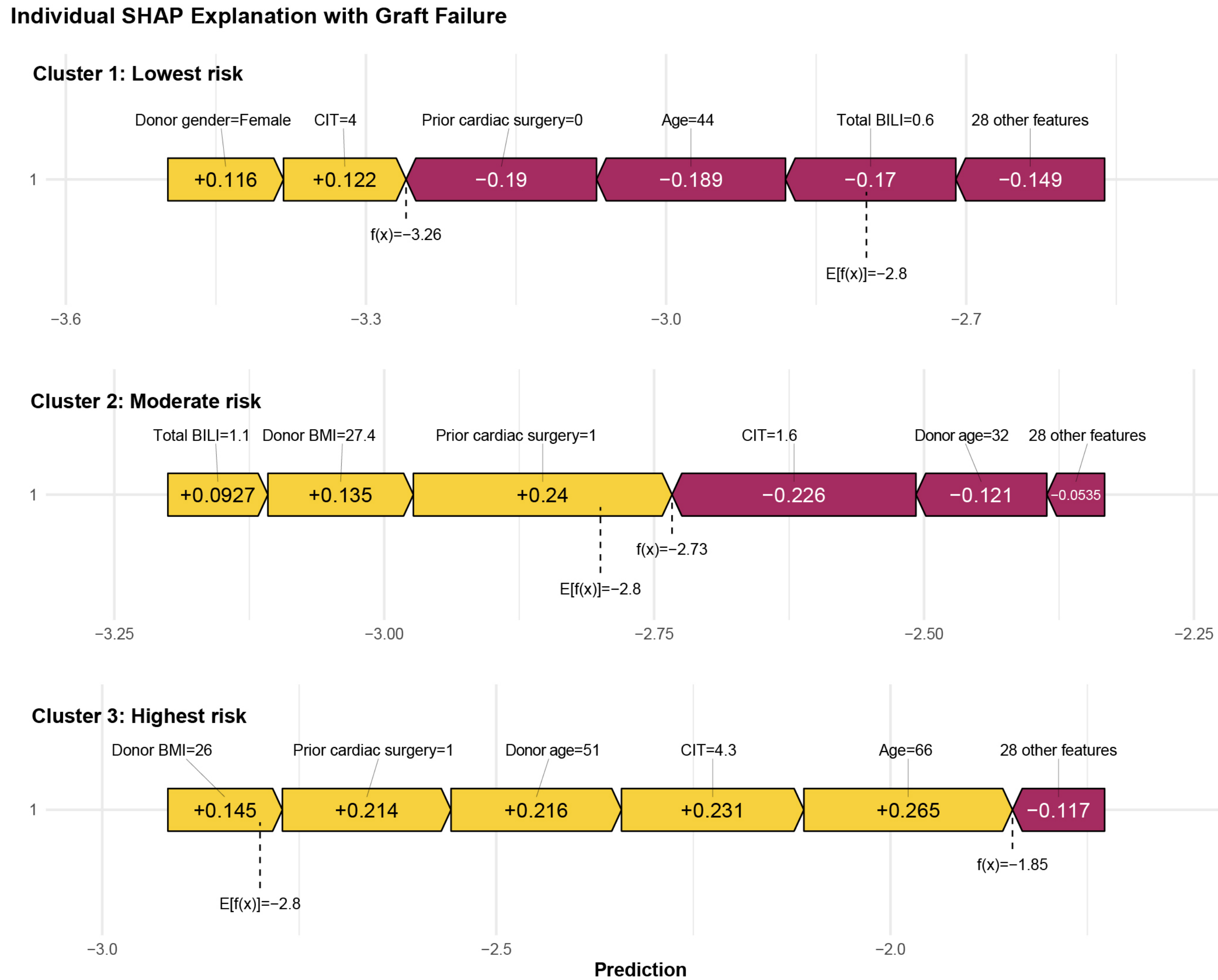

Figure 4 Patient-specific risk explanation using SHAP force plots. Three representative cases illustrate the local interpretability of the XGBoost model using SHAP force plots. Cases are selected from the low-risk (top), moderate-risk (mid) and highest-risk (bottom) clusters identified in figure 3. This demonstrates how feature contributions diverge between risk clusters. BMI, body mass index; CIT, cold ischaemia time; SHAP, SHapley Additive exPlanations; BILI, total bilirubin.

group demonstrated a significantly higher incidence of continuous renal replacement therapy (CRRT) (23% vs 16%, p=0.048). Extracorporeal membrane oxygenation (ECMO) use was numerically more frequent in the high-risk cohort (9.0% vs 5.0%, p=0.10), though this difference did not meet the threshold for statistical significance. Rates of acute rejection (1.1% vs 0.4%, p=0.60) and intra-aortic balloon pump utilisation (33% vs 28%, p=0.21) did not differ significantly between groups. Overall, these findings highlight CRRT as a critical driver of post-transplant morbidity in the high-risk cohort, with ECMO showing a clinically notable trend warranting further investigation.

DISCUSSION

This study demonstrates how ML can refine EGF risk prediction by screening for and prioritising interactions between donor, recipient and procedural factors. We acknowledge that the final nomogram, being a multivariate logistic regression model, is inherently additive and does not explicitly incorporate the non-linear interaction terms identified during the XGBoost discovery phase. However, the XGBoost-derived feature selection ensured that only the most robust predictors—those maintaining impact even across potential non-linearities—were included in the final clinical tool. Our XGBoost-derived nomogram achieved moderate discrimination (AUC: 0.67)—an AUROC performance that was effectively maintained when translating from complex ensemble models to a simpler logistic regression framework (ΔAUC=0.007). This similarity in performance suggests that for 90-day graft failure, the majority of the predictive signal is captured by the linear combination of these high-importance variables. Furthermore, the model significantly outperforms traditional benchmarks such as the IMPACT score (AUC: 0.586 in our

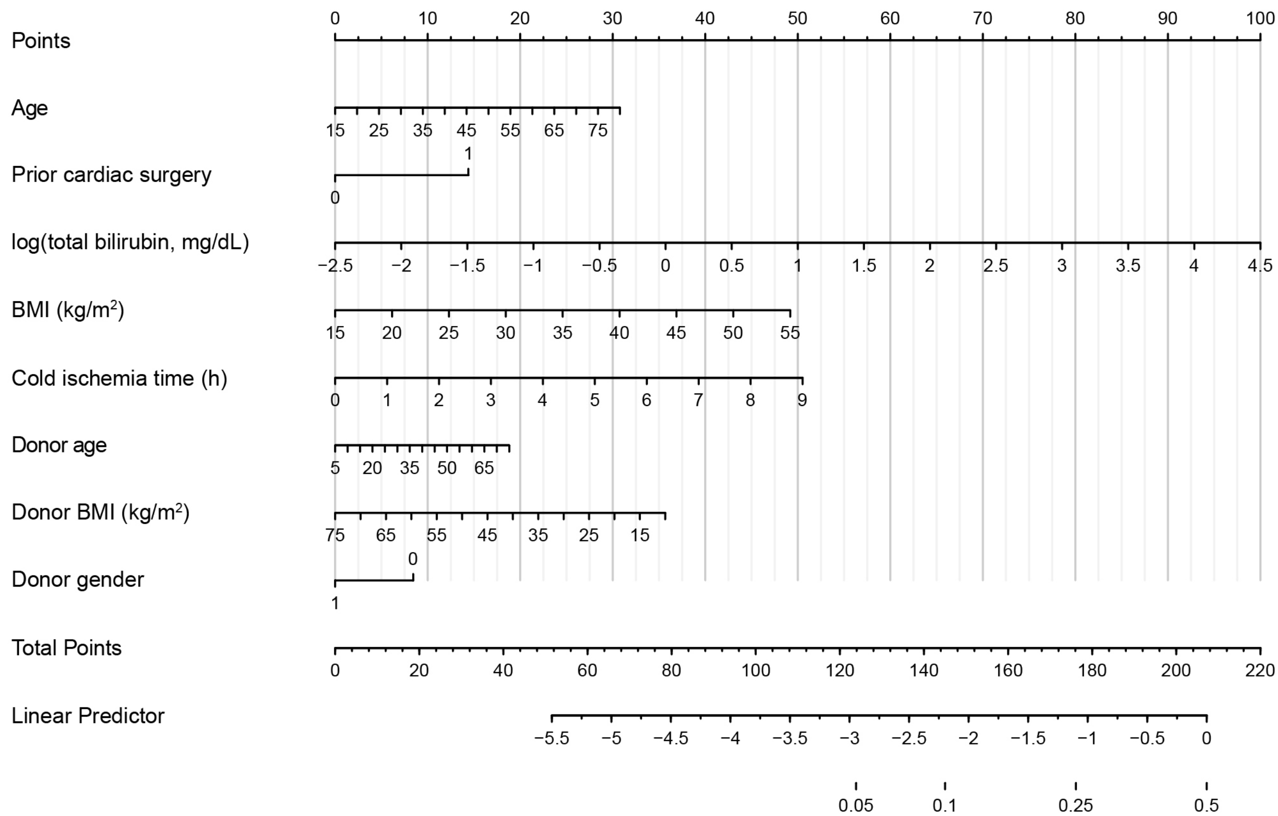

Figure 5 Clinical nomogram derived from ML-prioritised predictors. A clinically interpretable nomogram constructed via logistic regression using eight high-importance predictors identified through global SHAP importance rankings. This clinically interpretable nomogram translates machine learning-identified risk factors into actionable bedside predictions for 90-day graft failure. Scoring scales reflecting their nonlinear contributions. BMI, body mass index; ML, machine learning; SHAP, SHapley Additive exPlanations.

head-to-head comparison) for the specific prediction of 90-day outcomes. By maintaining clinical interpretability through SHAP visualisation, this tool bridges the 'black-box' translational gap often associated with artificial intelligence (AI) in medicine.

A major limitation of traditional tools like the IMPACT score is their development based solely on recipient factors, intentionally excluding donor-specific and procedural variables to maintain simplicity. However, graft survival is a product of complex interactions across the entire transplant process. By incorporating eight multidimensional predictors—including recipient factors (prior surgery, age, bilirubin, BMI), donor factors (age, gender, BMI) and CIT—our model provides a more holistic and accurate assessment of risk. Traditional scores rely on multivariable logistic regression, which assumes linear relationships and often overlooks how variables interact. Crucially, external validation in a Chinese cohort confirmed cross-continental generalisability, addressing a key gap in transplantation AI models.

The identified predictors align with established biological pathways. Prior cardiac surgery reflects technical complexity and systemic vulnerability, while elevated bilirubin indicates hepatic dysfunction from right heart failure. Prolonged CIT directly impacts myocardial viability, and donor factors (age, BMI) correlate with subclinical cardiac abnormalities.[21 22] Centre volume embodies institutional expertise in managing these cumulative risks.

SHAP analysis revealed critical non-linear interactions, such as compounded risk from older donor age with prolonged ischaemia—a dynamic poorly captured by traditional regression. This granularity enables refined donor–recipient matching, particularly for high-risk subgroups where targeted strategies (eg, prioritising younger donors) may mitigate injury pathways.

While the model's moderate AUC reflects EGF's inherent complexity, its clinical value lies in stratifying patients into distinct risk categories. High-risk patients experienced 17% graft failure incidence—more than double the low-risk group. Several limitations should be considered. First, our model is intentionally built on pretransplant variables available in large registries, excluding perioperative parameters (eg, donor troponin, postreperfusion haemodynamics) and granular immunological data. While this ensures utility at the time of donor acceptance, it necessarily limits predictive granularity; future models integrating such dynamic data could enhance precision for postoperative management. Second, while external validation in a geographically distinct Chinese cohort supports generalisability, the

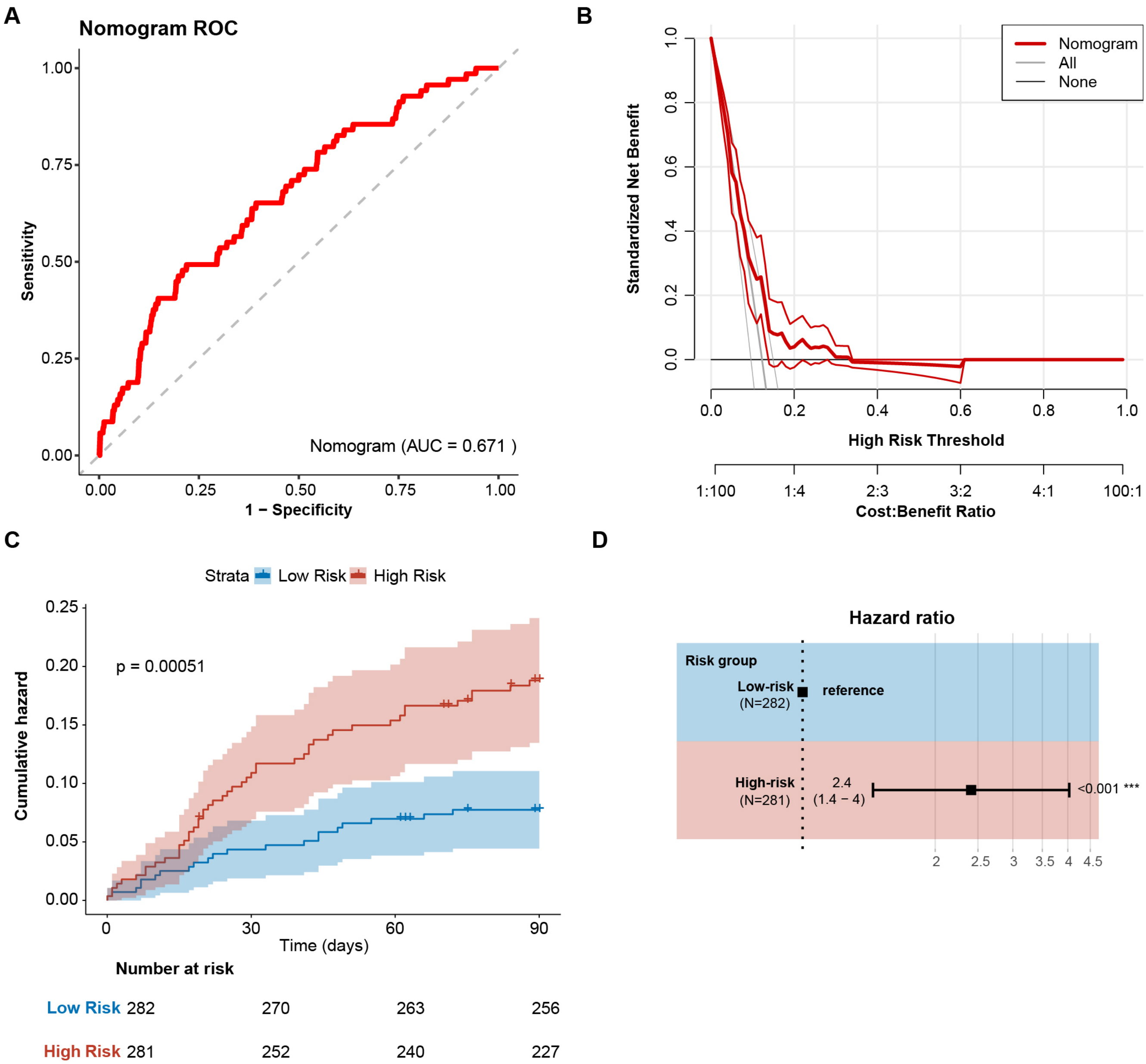

Figure 6 External validation of the machine learning-driven nomogram in WHUH cohort. (A) Receiver operating characteristic (ROC) curve of the nomogram in the external validation cohort (N=563). The dotted diagonal line represents the reference for no discrimination. (B) Decision curve analysis (DCA) demonstrating clinical utility across risk thresholds. The nomogram (red line) outperformed 'treat all' (grey) and 'treat none' (black line) strategies. (C) Cumulative hazard curves stratified by nomogram-predicted risk (low-risk: score <30, blue; high-risk: score ≥30, red). Log rank test, p<0.05 as statistically significant. (D) Cox proportional hazards model forest plot confirming a 2.4-fold increased risk of graft failure in the high-risk group (HR=2.4; 95% CI: 1.4 to 4.0; p<0.001). The vertical dashed line marks the null effect (HR=1). Risk stratification was based on nomogram score thresholds derived from SHAP analysis. AUC, area under the curve; SHAP, SHapley Additive exPlanations; WHUH, Wuhan Union Hospital.

WHUH cohort is from a single centre and differences in clinical management may affect the calibration of absolute risk estimates, suggesting local recalibration may be warranted prior to implementation. Finally, the translation of the XGBoost model into a simpler nomogram entailed a minimal, non-significant loss in discrimination (ΔAUC=0.0049), which we believe is a favourable trade-off for achieving clinical interpretability and ease of use. Future work should integrate dynamic post-transplant biomarkers and prospectively evaluate nomogram-guided care.

CONCLUSION

In summary, this study demonstrates a pragmatic, two-stage pipeline for developing a clinical prediction tool. We employed explainable ML as a high-dimensional discovery engine to screen pretransplant variables and

Table 3 Donor and recipient baseline characteristics of the WHUH cohort with low and high risk

	Low risk, N=282*	High risk, N=281*	P value†
Age	46 (34, 54)	52 (44, 59)	<0.001
Male gender	214 (76)	223 (79)	0.4
Main diagnosis			0.001
NICM	200 (71)	155 (55)	
ICM	54 (19)	77 (27)	
VHD	14 (5.0)	33 (12)	
CHD	9 (3.2)	8 (2.8)	
Other	5 (1.8)	8 (2.8)	
BMI (kg/m^2)	22.5 (19.6, 24.7)	23.4 (20.8, 26.0)	<0.001
Prior cardiac surgery	34 (12)	119 (42)	<0.001
Donor age	37 (26, 45)	39 (28, 46)	0.3
Donor male	253 (90)	237 (84)	0.076
Donor BMI (kg/m^2)	22.49 (20.76, 24.22)	22.49 (20.76, 23.88)	>0.9
CIT (hour)	4.98 (2.42, 5.63)	6.10 (5.40, 6.80)	<0.001
Total bilirubin (mg/dL)	0.86 (0.63, 1.25)	1.54 (1.02, 2.53)	<0.001

*Median (IQR); n (%).
†One-way ANOVA; Pearson's χ^2 test.
BMI, body mass index; CHD, congenital heart disease; CIT, cold ischaemia time; ICM, ischaemic cardiomyopathy; NICM, non-ischaemic cardiomyopathy; VHD, valvular heart disease; WHUH, Wuhan Union Hospital.

prioritise eight robust predictors of 90-day graft failure. Translating these predictors into a simple, linear nomogram resulted in a tool with modest but consistent discrimination (AUC 0.67) that was successfully validated across US and Chinese cohorts.

While the final nomogram's performance is comparable to that of the complex ML model from which it was derived, the value of the approach lies in its process: using ML not as a black-box predictor, but as a rigorous method for variable screening and interaction analysis to inform a clinically interpretable model. This ML-informed nomogram outperforms traditional, recipient-centric scores for early risk prediction and offers a practical, validated aid for risk assessment at the critical point of donor acceptance, representing a tangible step towards generalisable, data-driven decision support in HTx.

Table 4 Post-transplant adverse events of low and high-risk group in WHUH Cohort

	Low risk, N=282*	High risk, N=281*	P value†
Acute rejection	1 (0.4)	3 (1.1)	0.6
CRRT	44 (16)	63 (23)	0.048
ECMO	14 (5.0)	25 (9.0)	0.10
IABP	77 (28)	93 (33)	0.2

*Median (IQR); n (%).
†One-way ANOVA; Pearson's χ^2 test.
ANOVA, analysis of variance; CRRT, continuous renal replacement therapy; ECMO, extracorporeal membrane oxygenation; IABP, intra-aortic balloon pump; WHUH, Wuhan Union Hospital.

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Patient consent for publication Not applicable.

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Data availability statement Data are available upon reasonable request. The data used in this study are derived from two distinct sources with varying access requirements: UNOS/OPTN Data: The primary derivation data were supplied by the United Network for Organ Sharing (UNOS) as the contractor for the Organ Procurement and Transplantation Network (OPTN). These data are available upon request to the OPTN through their official website (https://optn.transplant.hrsa.gov/data/view-data-reports/request-data/). Access is subject to the data use agreements and approval processes of the OPTN.WHUH Data: The external validation data from Wuhan Union Hospital (WHUH) are not publicly available due to institutional ethical restrictions and patient privacy regulations. These data are maintained in-house and are restricted to authorised personnel involved in the study; as such, they cannot be shared or distributed externally.

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